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The effect of geometric parameters on mixed convection in an inclined eccentric annulus

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ABSTRACT

Laminar mixed convection in an inclined eccentric straight annulus is studied numerically in this article. The coupled Navier–Stokes and energy equations are solved using finite difference method. The projection algorithm is employed for solution and to cope with the problem geometry, the cylindrical bipolar coordinate is used. To model natural convection, Boussinesq approximation is used for the body force term in the equations of motion. Developing hydrodynamic and thermal conditions are assumed for simulation. The inner wall is subjected to a fixed heat flux and the outer one is insulated. The effect of different parameters such as eccentricity, radius ratio and inclination on flow field and thermal characteristics is studied. It has been found that the Nusselt number has a local maximum value about 55° from vertical by laying down the annuli.

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1. Introduction

Mixed convection in an inclined eccentric straight pipe is encountered in many engineering problems such as heat exchangers, directional drilling, chemical reactors, cooling of electric motors, thermal energy storages, solar energy collectors etc. Therefore, understanding the flow field and its thermal characteristics will help engineers to design more efficient systems.

Laminar forced convection in eccentric straight annulus has been studied by some researchers including both the fully developed and developing flows. Reynolds et al. [1] and Heaton et al. [2] studied flow field and heat transfer in an eccentric annulus for fully developed flow with four different kinds of thermal boundary conditions. They used an analytical approach for the equations of motion and a numerical solution for the energy equation. In another work, a solution for temperature distribution for fluid flowing in an eccentric annulus with the assumption of fully developed temperature profile and determination of heat transfer convection has been presented by Snyder [3], who studied the heat transfer and its coefficient in this problem. His conclusion was that the average Nusselt number decreases as eccentricity increases. Cheng and Hwang [4] focused on an analytical approach to solve the energy equation in eccentric annuli under fully developed assumption. Siginer and Bakhtiyarov [5] studied the effect of pipe eccentricity on flow of well bore in annuli both analytically and experimentally. They developed an analytical expression for azimuthal velocity as a function of eccentricity and rheological properties of the fluid. Also, velocity profiles were measured for Newtonian and non-Newtonian fluids numerically. Trombetta [6] investigated laminar forced convection in eccentric annuli with different thermal boundary conditions using analytical and numerical approaches. He concluded that the average Nusselt number decreases as radius ratio increases, with the inner wall subjected to constant heat flux and the outer one insulated. Feldman et al. [7,8] studied flow field and temperature profile in an eccentric annulus under developing flow condition. They showed that hydrodynamic and thermal entrance lengths increase as eccentricity increases and with higher Prandtl number, the bulk temperature and heat transfer decrease. Also the variation of the Nusselt number along the annuli has been shown for both inner and outer walls. Nobari and Mehrabani [9] investigated the fluid flow and heat transfer in an eccentric curved annulus. They concluded that the Nusselt number can be increased at large values of Dean Number. Hosseini et al. [10] performed experimental study of turbulent forced convection in vertical eccentric annulus. They showed that the Nusselt number is related to the Reynolds number with a power correlation.

Laminar natural convection in a straight concentric and eccentric annulus also has been studied by some researchers numerically

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Nomenclature

a	the pole of bipolar coordinate system on the x axis (m)
B	body force (N)
c_p	specific heat at constant pressure (kJ/kg K)
D_h	hydraulic diameter, $2(r_o - r_i)$ (m)
e	eccentricity (m)
E	dimensionless eccentricity
g	physical gravitational acceleration (m/s^2)
g_η	gravitational acceleration in η direction (m/s^2)
g_ξ	gravitational acceleration in ξ direction (m/s^2)
g_z	gravitational acceleration in z direction (m/s^2)
Gr	Grashof number, $\beta g q' D_h^4/\nu^2 k$
h	coordinate scale factor (m)
H	dimensionless coordinate scale factor, h/D_h
h_η	coordinate scale factor in η direction (m)
h_ξ	coordinate scale factor in ξ direction (m)
l	length of channel (m)
k	thermal conductivity (W/m K)
$\vec{n}$	unit normal vector
N	radius ratio
Nu	average Nusselt number
Nu_i	circumferential local Nusselt number
Nu_{ic}	circumferential Nusselt number
P	pressure (Pa)
Pr	Prandtl number
q'	heat flux (W/m^2)
r_i	inner pipe radius (m)
r_o	outer pipe radius (m)
Re	Reynolds number, $w_m D_h/\nu$
t	time (s)
T	temperature (K)
T_e	inlet temperature (K)
T_{ic}	circumferential temperature (K)
T_m	mean temperature (K)
u	velocity component in ξ direction (m/s)
U	dimensionless velocity component in ξ direction, u/w_m
v	velocity component in η direction (m/s)
V	dimensionless velocity component in η direction, v/w_m
$\vec{V}$	dimensionless velocity vector
w	velocity component in z direction (m/s)
w_m	mean axial velocity (m/s)
W	dimensionless velocity component in z direction, w/w_m
Z	dimensionless length, z/l

Greek letters

α	thermal diffusivity (m^2/s)
β	volumetric coefficient of thermal expansion (1/K)
η	first bipolar coordinate (m)
θ	dimensionless temperature, $T - T_e/q' D_h \big/ k$
θ_{ic}	circumferential dimensionless temperature
θ_m	mean dimensionless temperature
μ	viscosity of fluid (kg/m s)
ν	kinematic viscosity (m^2/s)
ξ	second bipolar coordinate
ρ	density of fluid (kg/m^3)
ρ_e	density of fluid at ambient temperature (kg/m^3)
φ	inclination of annulus from axis in z direction (°)

Subscripts

c	circumferential
e	entrance
i	inner wall
m	average value
o	outer wall

but just in horizontal and vertical cases. Ho et al. [11] studied the effect of eccentricity, Rayleigh and Prandtl numbers on natural convection heat transfer in horizontal eccentric annuli. They showed that the Nusselt number mainly depends on Rayleigh number and eccentricity but changes slightly with the Prandtl number. El-Shaarawi and Mokheimer [12] investigated induced flow in an eccentric vertical annulus due to natural convection with inner wall subjected to constant heat flux and the outer one maintained at ambient temperature and vice versa. Mokheimer and El-Shaarawi [13] studied the effect of eccentricity and radius ratio on thermal characteristics of natural convection heat transfer in a vertical eccentric annulus. They considered fully developed flow condition with inner wall subjected to constant heat flux and the other outer maintained at ambient temperature and vice versa. The laminar, fully developed mixed convection in vertical eccentric annuli has been investigated by Sathyamurthy et al. [14]. They studied the effect of eccentricity and Rayleigh number on the friction factor and the Nusselt number. In another work, Choudhury and Karki [15] focused on mixed convection in a horizontal eccentric annulus to study the effect of radius ratio, eccentricity and Rayleigh number on the Nusselt number and flow field. Hirose et al. [16] performed numerical and experimental study on natural convection heat transfer in concentric and eccentric horizontal annuli. In their study, the outer elliptic tube is heated and the inner circular tube is cooled. The orientation of outer tube and eccentricity is varied as parameters. They concluded that increasing the eccentricity changes the heat transfer in various ways, depending on the angles of orientation. Also they observed that the local Nusselt number remains almost constant at various orientation angles. Naylor et al. [17] investigated the natural convective heat transfer in a horizontal annulus by numerical and experimental techniques. The inner and outer walls of the annulus are maintained at different temperatures. They studied the effect of angular position of inner pipe and the eccentricity on the heat transfer. Hosseini et al. [18] studied the heat transfer in an open-ended vertical eccentric annulus experimentally with the outer wall subjected to a constant heat flux and the inner one wall being insulated. They showed that the Nusselt number can be correlated to Rayleigh number using a power function. All the above studies are focused on the vertical or horizontal eccentric pipe.

In the present work, natural convective heat transfer in an inclined eccentric annulus has been studied. Using the second order finite difference numerical solution technique, laminar steady state flow is modeled. The velocity and temperature profiles are assumed to be, developing. The inner wall is subject to constant heat flux and the outer wall is insulated. Fluid flow and temperature variation in the annulus is investigated and the effect of non-dimensional parameters such as eccentricity, radius ratio and inclination angle is studied. Variations in axial flow, secondary flow pattern, bulk temperature, inner wall temperature and Nusselt number are investigated in detail.

The physics of problem including the physical and computational domains and their figures are described in Section 2. Governing equations comprising continuity, momentum and energy equations, and hydrodynamic and thermal boundary conditions are presented in Section 3. After introducing the Projection algorithm

as numerical method briefly in Section 4, the accuracy of numerical code and validation of its results are surveyed in Section 5. Moreover, results are discussed in Section 6 and the summary and conclusions are presented in Section 7.

2. Problem description

The geometry of the problem consists of an outer tube with radius r_o and an inner tube with radius r_i. The tubes are aligned along the z axis and the centers of cross sections lay on the x axis, with a distance e from each other. The z axis makes an angle φ with the direction of the acceleration of gravity, g (Fig. 1). The fluid flows in the annulus region, between the tubes.

The cross sections of the tubes are shown in Fig. 2. Using the bipolar transformation, the annulus region can be mapped into a rectangle, with x and y coordinates transformed into ξ and η coordinates respectively (Fig. 2). A conformal structured grid can be generated by dividing this rectangle with regular subdivisions along the sides. Since the geometry of the transformed region is a simple rectangle, by applying the same mapping to the governing equations, and using the finite difference method, the problem can be solved in the computational domain.

3. Governing equations

In the present work the flow of a viscous fluid through the annulus is modeled. Transient effects are neglected and the steady state behavior is of interest. The equations governing the fluid flow and heat transfer are the continuity (Eq. (1)), the momentum (Eqs. (2)–(4)) and the energy (Eq. (5)) equations. These equations are derived in the cylindrical bipolar coordinate; The momentum equation (Eqs. (2)–(4)) is written in the ξ, η and z directions respectively.

$$\frac{1}{h}\frac{\partial u}{\partial \xi}+\frac{u}{h^2}\frac{\partial h}{\partial \xi}+\frac{1}{h}\frac{\partial v}{\partial \eta}+\frac{v}{h^2}\frac{\partial h}{\partial \eta}+\frac{\partial w}{\partial z}=0 \tag{1}$$

$$\begin{aligned}&\rho\frac{\partial u}{\partial t}+\rho\left(\frac{u}{h}\frac{\partial u}{\partial \xi}+\frac{v}{h}\frac{\partial u}{\partial \eta}+w\frac{\partial u}{\partial z}+\frac{uv}{h^2}\frac{\partial h}{\partial \eta}-\frac{v^2}{h^2}\frac{\partial h}{\partial \xi}\right)\\&=\mu\left(\frac{1}{h^2}\frac{\partial^2 u}{\partial \xi^2}+\frac{1}{h^2}\frac{\partial^2 u}{\partial \eta^2}+\frac{\partial^2 u}{\partial z}+\frac{2}{h^3}\frac{\partial v}{\partial \xi}\frac{\partial h}{\partial \eta}-\frac{u}{h^4}\left(\frac{\partial h}{\partial \eta}\right)^2\right.\\&\left.\quad-\frac{u}{h^4}\left(\frac{\partial h}{\partial \xi}\right)^2-\frac{2}{h^3}\frac{\partial v}{\partial \eta}\frac{\partial h}{\partial \xi}\right)-\frac{1}{h}\frac{\partial p}{\partial \xi}+B_\xi\end{aligned} \tag{2}$$

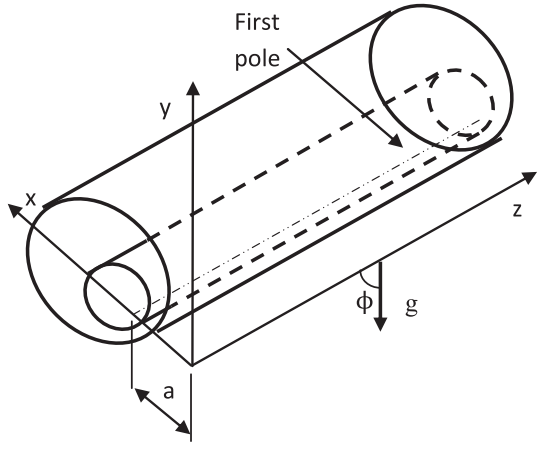


Fig. 1. Geometry of physical domain.

$$\begin{aligned}&\rho\frac{\partial v}{\partial t}+\rho\left(\frac{u}{h}\frac{\partial v}{\partial \xi}+\frac{v}{h}\frac{\partial v}{\partial \eta}+w\frac{\partial v}{\partial z}+\frac{uv}{h^2}\frac{\partial h}{\partial \xi}-\frac{u^2}{h^2}\frac{\partial h}{\partial \eta}\right)\\&=\mu\left(\frac{1}{h^2}\frac{\partial^2 v}{\partial \eta^2}+\frac{1}{h^2}\frac{\partial^2 v}{\partial \xi^2}+\frac{1}{h^2}\frac{\partial^2 v}{\partial \xi^2}-\frac{2}{h^2}\frac{\partial u}{\partial \xi}\frac{\partial h}{\partial \eta}+\frac{2}{h^3}\frac{\partial u}{\partial \eta}\frac{\partial h}{\partial \xi}\right.\\&\left.\quad-\frac{v}{h^4}\left(\frac{\partial h}{\partial \xi}\right)^2-\frac{v}{h^4}\left(\frac{\partial h}{\partial \eta}\right)^2\right)-\frac{1}{h}\frac{\partial p}{\partial v}+B_\eta\end{aligned} \tag{3}$$

$$\begin{aligned}&\rho\frac{\partial w}{\partial t}+\rho\left(\frac{u}{h}\frac{\partial w}{\partial \xi}+\frac{v}{h}\frac{\partial w}{\partial \eta}+w\frac{\partial w}{\partial z}\right)\\&=\mu\left(\frac{\partial^2 w}{\partial z^2}+\frac{1}{h^2}\frac{\partial^2 w}{\partial \xi^2}+\frac{1}{h^2}\frac{\partial^2 w}{\partial \eta^2}\right)-\frac{\partial p}{\partial z}+B_z\end{aligned} \tag{4}$$

$$\rho c_p\left(\frac{\partial T}{\partial t}+\frac{u}{h}\frac{\partial T}{\partial \xi}+\frac{v}{h}\frac{\partial T}{\partial \eta}+w\frac{\partial T}{\partial z}\right)=k\left(\frac{1}{h^2}\frac{\partial^2 T}{\partial \xi^2}+\frac{1}{h^2}\frac{\partial^2 T}{\partial \eta^2}+\frac{\partial^2 T}{\partial z^2}\right) \tag{5}$$

Parameter h in the above equations is the scale factor for the cylindrical bipolar coordinate system and is defined as:

$$\begin{aligned}h&=h_\xi=h_\eta=\frac{a}{\cosh\eta-\cos\xi}\\h_z&=1\end{aligned} \tag{6}$$

In which a is the x location of the pole of the cylindrical bipolar coordinate system and can be expressed as:

$$a=r_o\sinh\eta_o=r_i\sinh\eta_i \tag{7}$$

For certain annulus geometry, the transformation for the inner and outer walls can be defined as:

$$\eta_i=\cosh^{-1}\left(\frac{1}{N}\frac{N(1+E^2)+(1-E^2)}{2E}\right) \tag{8}$$

$$\eta_o=\cosh^{-1}\left(\frac{(1+E^2)+N(1-E^2)}{2E}\right) \tag{9}$$

In which E is the non-dimensional eccentricity and N is the ratio of the inner to outer radii.

$$E=\frac{e}{r_o-r_i} \tag{10}$$

$$N=\frac{r_i}{r_o} \tag{11}$$

The body force components, B_i, takes into account the weight of the fluid and can be expressed as:

$$B_\xi=\rho g_\xi \tag{12}$$

$$B_\eta=\rho g_\eta \tag{13}$$

$$B_z=\rho g_z \tag{14}$$

Components of the gravitational acceleration, g, in each direction are calculated using the following relations based on the transformation parameters and the inclination angle, φ, relative to the vertical direction.

$$g_\xi=g\frac{\cosh\eta\cos\xi-1}{a}\sin\varphi \tag{15}$$

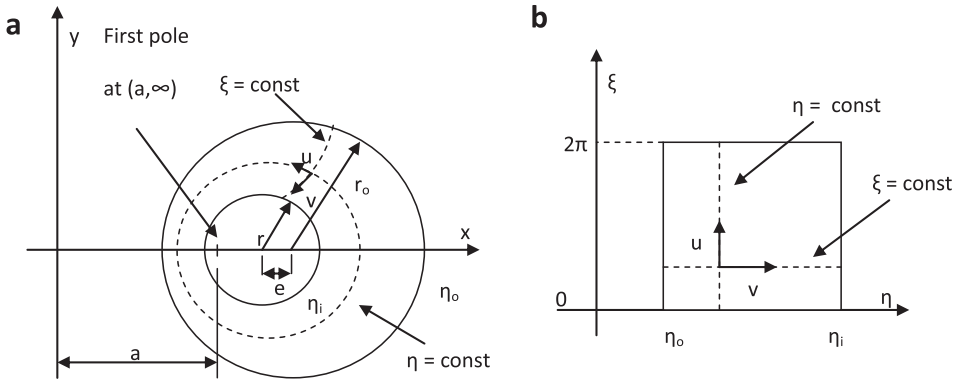


Fig. 2. Geometry in the bipolar coordinate: a: physical domain; b: computational domain.

$$g_\eta = g\frac{\sinh\eta\sin\xi}{a}\sin\varphi \tag{16}$$

$$g_z = g\cos\varphi \tag{17}$$

In the natural convection heat transfer, temperature difference in the fluid will cause density variations, which in turn causes body force gradient that will drive the fluid flow. The fluid density variation with temperature is implemented via the Boussinesq approximation. Therefore, the body force term becomes:

$$(\rho-\rho_e)g = -\rho_i g\beta(T-T_e) \tag{18}$$

where ρ and ρ_e are densities at T and T_e (the inlet fluid temperature) respectively and β is the volumetric thermal expansion coefficient.

To complete the mathematical model, suitable boundary conditions must be defined for the governing equations. Fluid enters the annulus from the bottom at ambient temperature and exits with a fully developed temperature profile from the top. Hence a uniform inlet temperature (Eq. (19)) and a constant temperature gradient (Eq. (20)) are used as the boundary condition for inlet and outlet respectively. The inner wall is subjected to a constant heat flux and the outer wall is insulated. These conditions are expressed respectively by Eqs. (21) and (22) as:

$$T = T_i \text{ or } \theta_{z=0} = 0 \tag{19}$$

$$\left(\frac{\partial T}{\partial z}\right)_{z=l} = \left(\frac{\partial T_m}{\partial z}\right)_{z=l} = \frac{2q'r_i}{\rho w_m c_p\left(r_o^2-r_i^2\right)}$$
$$\left(\frac{\partial\theta}{\partial Z}\right)_{z=l} = \left(\frac{\partial\theta_m}{\partial Z}\right) = \frac{4N}{1+N}\frac{1}{Re\,Pr} \tag{20}$$

$$\left(\frac{\partial T}{\partial n}\right)_{\eta_i} = \frac{q'}{k}$$
$$\left(\frac{\partial\theta}{\partial\eta}\right)_{\eta_i} = H(\eta_i,\xi) \tag{21}$$

$$\left(\frac{\partial T}{\partial\eta}\right)_{\eta_o} = 0$$
$$\left(\frac{\partial\theta}{\partial\eta}\right)_{\eta_o} = 0 \tag{22}$$

From the hydrodynamic point of view, to impose the physical condition at inlet, the fluid enters with a constant uniform velocity and exits with a developed velocity profile. On the walls, the no slip condition applies. These conditions are implemented using Eqs. (23)–(25) respectively.

$$u_{z=0} = v_{z=0} = 0,\ w_{z=0} = w_m$$
$$U_{z=0} = V_{z=0} = 0,\ W_{z=0} = 1 \tag{23}$$

$$\left(\frac{\partial u}{\partial z}\right)_{z=l} = \left(\frac{\partial v}{\partial z}\right)_{z=l} = \left(\frac{\partial w}{\partial z}\right)_{z=l} = 0$$
$$\left(\frac{\partial U}{\partial Z}\right)_{z=l} = \left(\frac{\partial V}{\partial Z}\right)_{z=l} = \left(\frac{\partial W}{\partial Z}\right)_{z=l} = 0 \tag{24}$$

$$U_{\eta_o} = U_{\eta_i} = V_{\eta_i} = V_{\eta_o} = W_{\eta_i} = W_{\eta_o} = 0 \tag{25}$$

The circumferential average temperature on inner wall, the bulk temperature and the local Nusselt number on inner wall and their dimensionless form, for the boundary conditions considered here, can be determined, respectively, as:

$$T_{ic} = \frac{1}{2\pi r_i}\int_0^{2\pi}(Th)_{\eta=\eta_i}d\xi$$
$$\theta_{ic} = \frac{1-N}{N\pi}\int_0^{2\pi}(\theta H)_{\eta=\eta_i}d\xi \tag{26}$$

$$T_m = \frac{1}{\pi w_m\left(r_o^2-r_i^2\right)}\int_0^{2\pi}\int_{\eta_o}^{\eta_i}Twh^2d\eta\,d\xi$$
$$\theta_m = \frac{4(1-N)}{\pi w_m(1+N)}\int_0^{2\pi}\int_{\eta_o}^{\eta_i}\theta UH^2d\eta\,d\xi \tag{27}$$

Table 1
Grid independency test in three different mesh sizes of 30 × 20, 50 × 30 and 60 × 40 at $Gr = 1.35\times10^5$, $Re = 300$, $E = 0.5$, $N = 0.5$ and $\varphi = 45$.

Axial velocity contours			
Mesh size	30 × 20	50 × 30	60 × 40
Nusselt number	7.233851	7.303483	7.405874

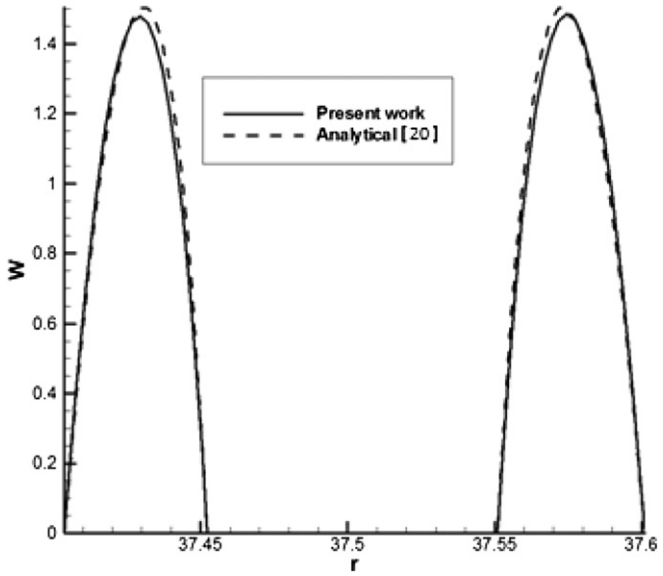


Fig. 3. Comparison of velocity profile of analytical solution [20] for concentric straight pipe with the present work at $E = 0.01$, $N = 0.5$.

$$Nu_{\mathrm{i}} = \frac{q' D_{\mathrm{h}}}{k\left(T_{\eta=\eta_{\mathrm{i}}} - T_{\mathrm{m}}\right)}$$
$$Nu_{\mathrm{i}} = \frac{1}{\theta_{\eta=\eta_{\mathrm{i}}} - \theta_{\mathrm{m}}} \tag{28}$$

Due to insulation on the outer wall, the Nusselt number is zero and, its temperature is not studied here. The circumferential local Nusselt number and the total Nusselt number on the inner wall can be expressed by Eqs. (29) and (30) respectively.

$$Nu_{\mathrm{ic}} = \frac{q^{'} D_{\mathrm{h}}}{k(T_{\mathrm{ic}} - T_{\mathrm{m}})}$$
$$Nu_{\mathrm{ic}} = \frac{1}{\theta_{\mathrm{ic}} - \theta_{\mathrm{m}}} \tag{29}$$

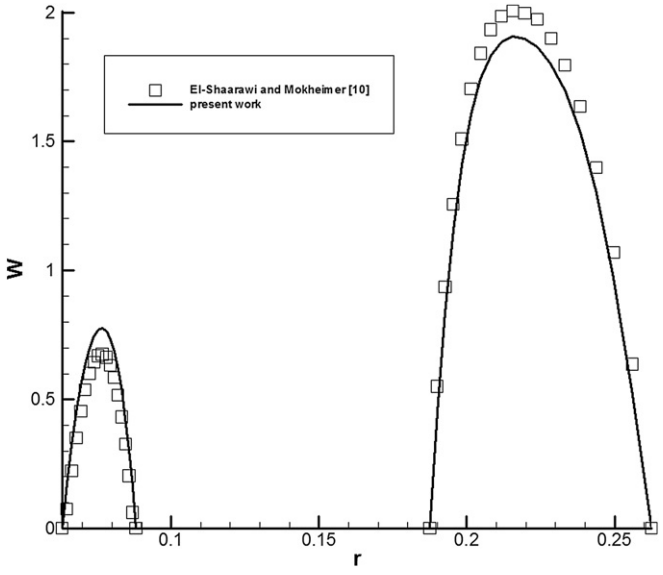


Fig. 4. Comparison of velocity profile of the El-Shaarawi and Mokheimer work [12] with the present work at $E = 0.5$ and $N = 0.5$.

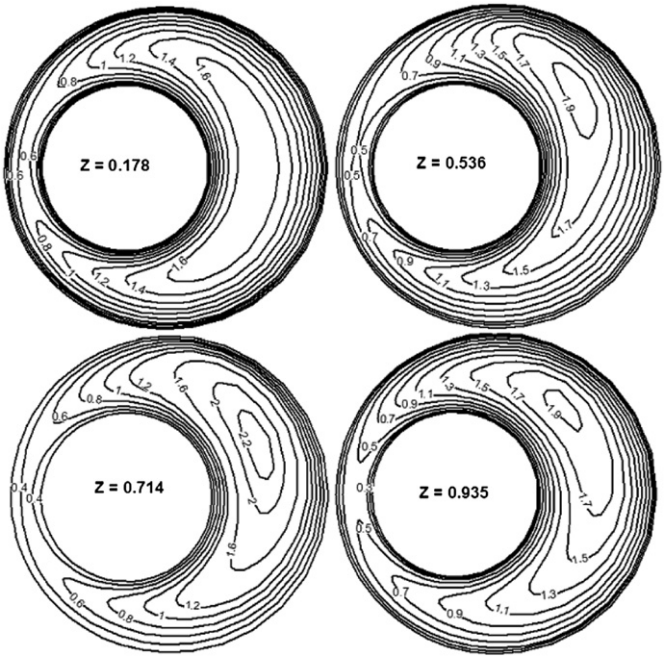


Fig. 5. Developing axial velocity contour lines at $N = 0.5$, $E = 0.5$ and $\varphi = 45$.

$$Nu = \frac{1}{l}\int_0^l Nu_{\mathrm{ic}}\mathrm{d}z$$
$$Nu = \int_0^1 Nu_{\mathrm{ic}}\mathrm{d}Z \tag{30}$$

4. Numerical method

To solve the governing equations, the projection algorithm [19] is employed. The second order differencing method used is forward

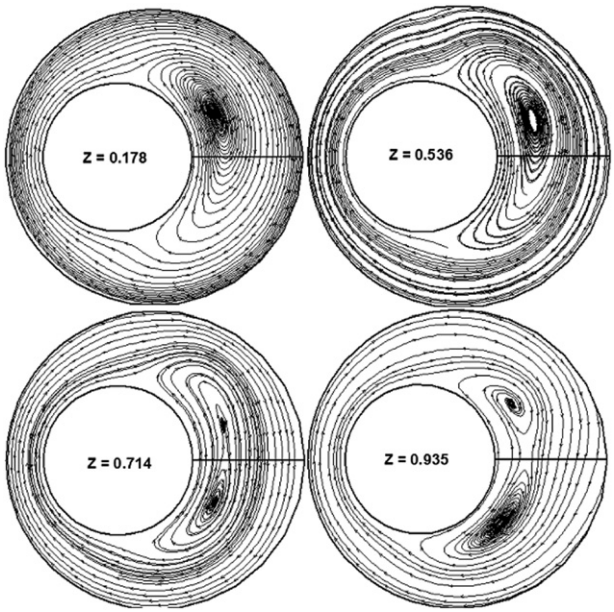


Fig. 6. Developing streamlines at $N = 0.5$, $E = 0.5$ and $\varphi = 45$.

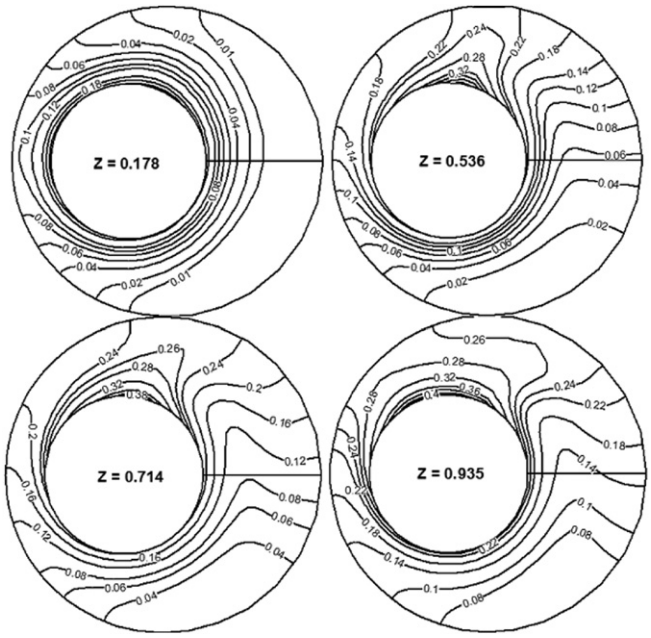


Fig. 7. Developing temperature contour lines at $N = 0.5$, $E = 0.5$ and $\varphi = 45$.

in time and central in space. To obtain the steady state solution, a time marching strategy for the numerical scheme is employed. The steady state condition is reached when the time derivative terms in the momentum and energy equations approach zero. Hence, the calculation stopping condition can be specified as follows:

$$\left(\max\left|\frac{\partial u}{\partial t}\right|, \max\left|\frac{\partial v}{\partial t}\right|, \max\left|\frac{\partial w}{\partial t}\right|, \max\left|\frac{\partial T}{\partial t}\right|\right) \le \varepsilon \tag{31}$$

In which ε is a small value close to zero depending on accuracy of the solution. Here, it is taken equal to 10^{-3}. Equations (32) and (33) are obtained by splitting the momentum equation, with the introduction of the auxiliary velocity, V^*. The continuity equation written in the time discretized case becomes Eq. (34), as the calculated velocities in the next time step should be divergence-free. Applying this condition on Eq. (33), will result in Eq. (35), which is a Poisson equation for the pressure variable.

$$\frac{\vec{V}^* - \vec{V}^n}{\Delta t} + \left((\vec{V}\cdot\nabla)\cdot\vec{V}\right)^n = \nu\left(\nabla^2\vec{V}\right)^n \tag{32}$$

$$\frac{\vec{V}^{n+1} - \vec{V}^*}{\Delta t} + \frac{\nabla P^{n+1}}{\rho} = 0 \tag{33}$$

$$\nabla\cdot\vec{V}^{n+1} = 0 \tag{34}$$

$$\nabla^2 P^{n+1} = \frac{\rho}{\Delta t}\vec{V}^* \tag{35}$$

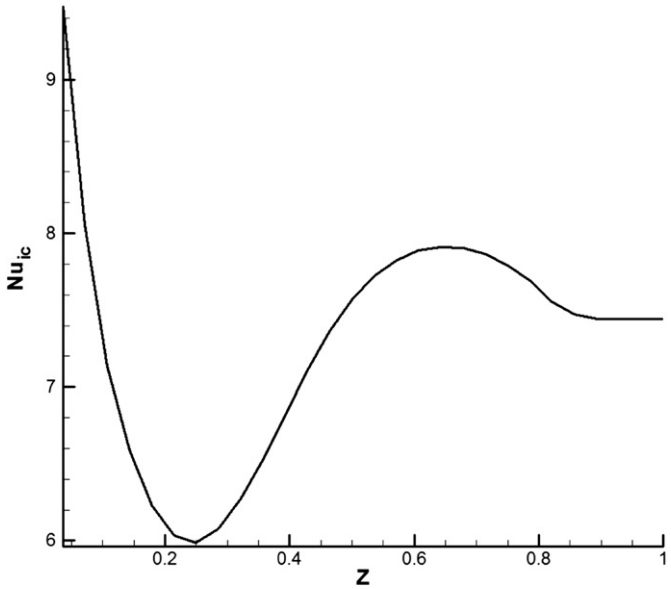


Fig. 8. Variation of the circumferential local Nusselt number at $N = 0.5$, $E = 0.5$ and $\varphi = 45$.

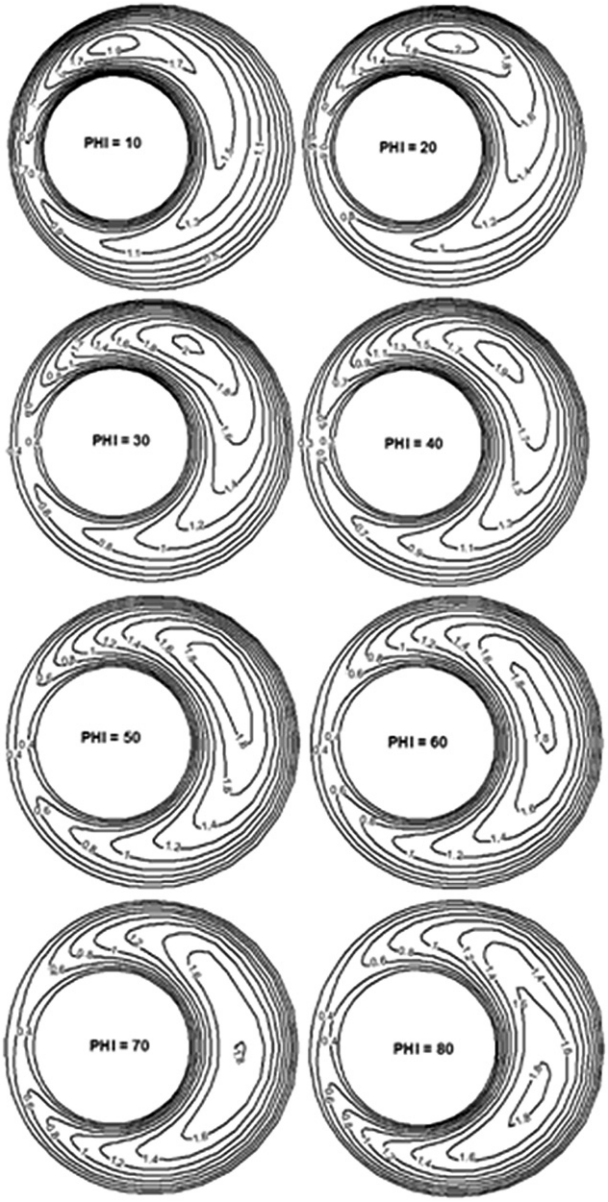


Fig. 9. Axial velocity contour lines at $E = 0.5$ and $N = 0.5$ for different inclination angles.

Based on the above projection, it can be shown that the pressure field should be independent of the boundary value of the auxiliary velocity, V^*. Hence, a Neumann condition for the pressure at all boundaries can be applied as in Eq. (36), which means that the value of V^* should equal to the physical velocity values on the boundaries.

$$\frac{\partial P}{\partial n} = 0 \tag{36}$$

Staggered grid is used to discretize the domain in η and in ξ directions. For the domain defined in bipolar-cylindrical coordinate system (Fig. 2), η varies between η_o and η_i and ξ varies between 0 and 2π. Hence, the computational domain shown in Fig. 2 is divided into a uniform space steps in η and ξ directions.

5. Grid independency test and code accuracy

To investigate the accuracy and conservative property of the numerical code which was developed, the simple case of concentric pipe is simulated using three different meshes with grid count of 30×20, 50×30 and 60×40 in the (ξ,η) direction. The number of axial mesh size varies from 30 to 80 depending on each geometric parameter in order to having hydrodynamic and thermal fully developed conditions at outlet. The numerical results have a maximum three percent deviation (Table 1); obviously indicating the grid independency of the code. Therefore, the 50×30 mesh has been chosen for the subsequent numerical simulations.

To check the accuracy of the code, the velocity profile obtained for a case with small value of eccentricity is compared to the analytical solution of the corresponding concentric pipe [20] Eq. (37), under forced flow assumption (Fig. 3).

$$U = \frac{2}{1 + N^2 - \dfrac{N^2 - 1}{\ln(N)}}\left[1 - \left(\frac{r}{r_o}\right)^2 + \frac{N^2 - 1}{\ln(N)}\ln\left(\frac{r}{r_o}\right)\right] \tag{37}$$

In a second test, the velocity profile calculated for an eccentric vertical pipe with natural convection assumption is compared with the El-Shaarawi and Mokheimer work [12] (Fig. 4).

6. Results and discussion

The physical domain is shown in Fig. 1. This domain is fitted with the bipolar cylindrical coordinate system indicated in Fig. 2. Also, due to the uniform orthogonal divisions in η and ξ directions as it is shown in Fig. 2b, it is convenient to use the finite difference scheme here. All the following results are obtained at $Re = 300$, $Gr = 1.35 \times 10^5$ and $Pr = 0.7$. It also has to be noted that each case has a different CPU time for convergence which is varied from 30 hours to 4 days based on its geometric condition.

Developing axial flow can be divided into four stages and indicates in Fig. 5, at $E = 0.5$, $N = 0.5$ and $\varphi = 45$. In the first stage, under the effect of the viscous force, axial flow profile is shaped while due to temperature gradient, inclines toward the inner wall. In the second stage, as temperature increases in the upper section of the annulus, the buoyancy force becomes stronger and the peak of axial velocity profile translates counterclockwise and increases. In the third stage, as the flow proceeds downstream, the

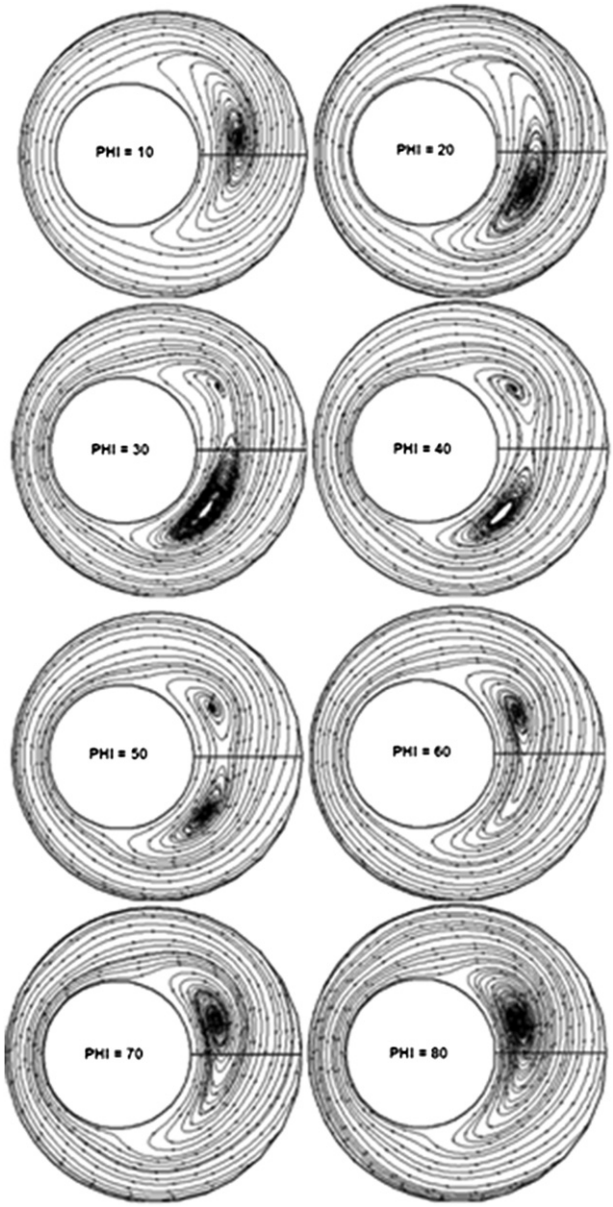


Fig. 10. Streamlines at $E = 0.5$ and $N = 0.5$ for different inclination angles.

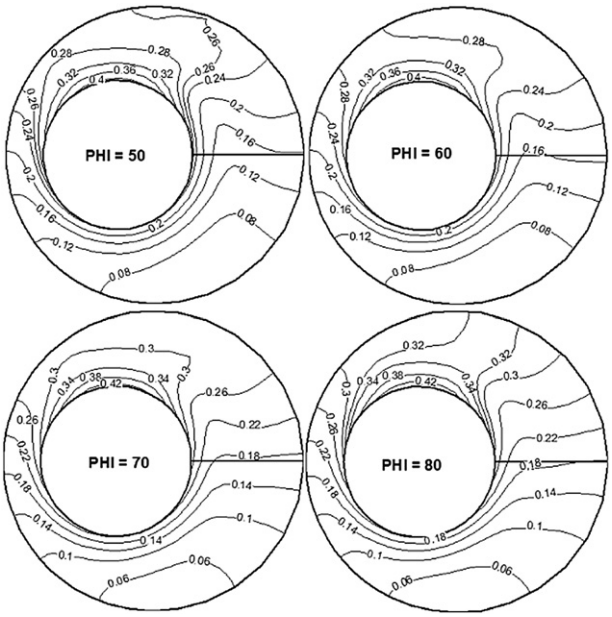


Fig. 11. Temperature contour lines at $E = 0.5$ and $N = 0.5$ for different inclination angles.

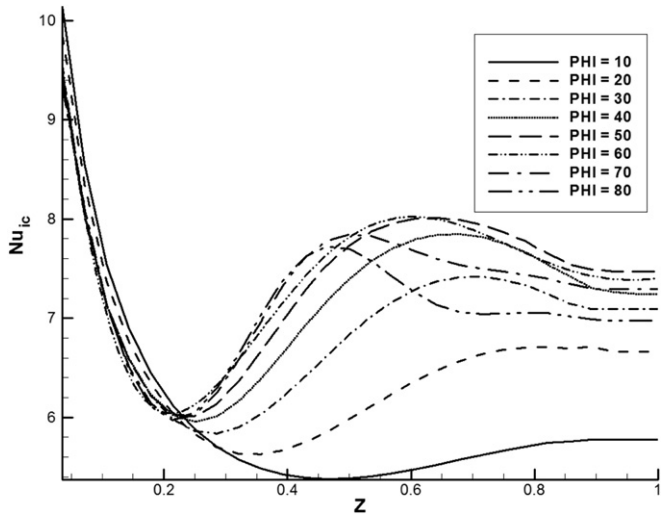


Fig. 12. Circumferential local Nusselt number of inner wall versus dimensionless length at $E = 0.5$ and $N = 0.5$ for different angle of inclination angles.

circumferential secondary flow becomes strong enough to push the maximum axial velocity clockwise. Also, the higher temperature increases the maximum axial velocity. In the fourth stage, heat is diffused to the outer and lower layers of the fluid, increasing the buoyancy force in those regions and the bulk velocity of the fluid. As a result, the maximum of the velocity profile is decreased.

The secondary flow field and temperature contours are shown in Figs. 6 and 7, respectively, at cross sections, corresponding to the previous figure. As it is evident, two secondary flows, a circumferential flow and a vortex flow exist. The driving force for the secondary flows is the high temperature gradient, near the inner wall. In the narrow section, the fluid is heated and goes up, gaining momentum and adhering to the outer wall. When the flow reaches the cold region, around the wide section, it loses heat but due to its momentum goes on and returns to the narrow section, creating a circumferential flow. Near the inner tube, on the right side, a secondary flow, also in the upward direction, is established. Under the effect of the circumferential flow, this new stream changes direction and creates the vortex flow. At $Z \leq 0.536$, bulk temperature is low; hence both circumferential and vortex flows are weak. As flow goes downstream ($Z > 0.536$) the bulk temperature rises, therefore secondary flows become stronger. Especially, the circumferential flow intensifies enough to split the vortex flow. Moving downstream, heat is diffused in the fluid, decreasing the temperature gradient in the upper section. This reduces the intensity of the upper vortex and increases the intensity of the lower vortex. The diffusion of heat explains the behavior of axial velocity and secondary flows at different stages.

Fig. 8 demonstrates the circumferential local Nusselt number versus dimensionless length at the same conditions of Fig. 5. Variation of Nusselt number can be divided into three stages. In the first stage ($0 \leq Z \leq 0.24$), due to the unformed thermal and hydrodynamic boundary layers and low velocity around the inner

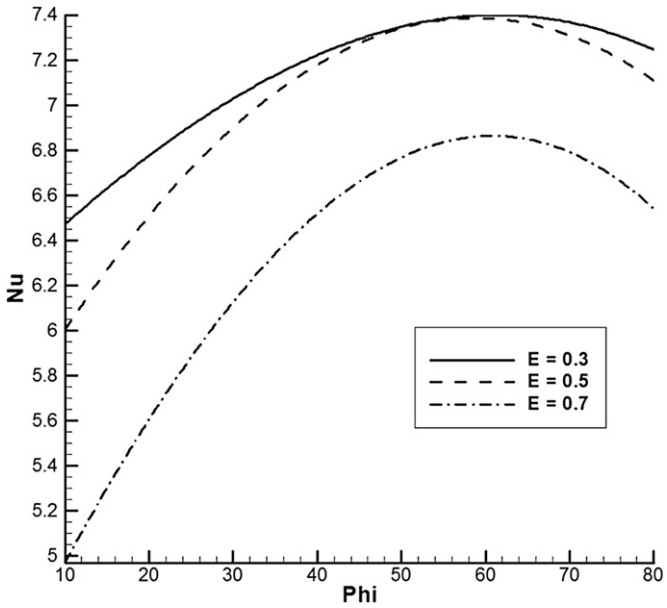


Fig. 13. Average Nusselt number of inner wall versus inclination angel at $N = 0.5$ for three different eccentricities.



Fig. 14. Axial velocity contour lines at $N = 0.5$ and $\varphi = 45$ for different eccentricities.

wall, temperature of the inner wall increases rapidly and the Nusselt number decreases. In the next stage ($0.24 < Z \leq 0.68$), the bulk temperature rises. Consequently, axial and secondary velocities around the inner wall increase so does the Nusselt number. In the third stage ($0.68 < Z \leq 0.91$), by diffusion of heat to outer layers of the fluid and reduction of axial velocity around the inner pipe, Nusselt number decreases.

Fig. 9 shows the axial velocity contours for different inclination angles, at cross sections where the velocity profile becomes fully developed. In near vertical annuli, the axial velocity profile is symmetric and inclines toward the inner wall. The secondary flows are negligible due to the buoyancy force being in the axial direction. Inclining the pipe, causes the buoyancy force to have a component in the cross sectional plane. At small degrees of inclination ($0 \leq \varphi < 30$), this component is small and therefore, the effect of secondary flows on axial velocity is not significant. Consequently the peak of axial velocity is located above the inner wall where the temperature is high. As the inclination angle increases ($30 \leq \varphi$) secondary flows become stronger due to the increased cross-sectional component of buoyancy force. This affects axial flow and pushes it clockwise. By transferring the peak of the axial velocity profile to the wide region, its maximum decreases. In Fig. 10, secondary flow fields at different angles of inclination are demonstrated. In a vertical annuli due to the symmetry of flow field there is no secondary flow. As the annulus is inclined, the secondary flow comes into action as described earlier. In the range of small inclination, ($0 < \varphi \leq 20$), there is only a single clockwise vortex which is created by the momentum of circumferential flow. Approaching the annulus toward horizon results in an increase in the cross sectional component of the buoyant force, increasing the driving force of the circumferential flow. For $30 \leq \varphi \leq 50$ the circumferential flow is strong enough to split the vortex and make the upper one weak. As the annulus is more laid down ($\varphi \geq 60$), the vortex in the lower section becomes weaker and diminishes in the limit of horizontal annulus. This can be explained from the fact that the increased component of the buoyant force will cause stronger vortex flow, overcoming the circumferential flow and unifying the vortices. The result is an increase in the temperature of fluid in the lower section, as it can be seen from the temperature contour lines in Fig. 11.

The circumferential local Nusselt number on the inner wall versus dimensionless length at different angles of inclination is shown in Fig. 12. As inclination angle increases ($45 \leq \varphi$), in the second stage of variation of Nusselt, secondary flows intensify. Meanwhile the area subjected to the axial velocity becomes larger therefore the circumferential Nusselt number rises. Also, since the secondary flow becomes stronger, the Nusselt number increases at initial lengths of the annulus. Therefore, the peak of the Nusselt number curve moves toward the entrance.

Fig. 13 indicates variation of the average Nusselt number versus angle of inclination at three different eccentricities. It has a maximum around $\varphi = 55$, which shows the combined effect of axial flow decrease and secondary flow increase, with increasing the inclination angle. As the eccentricity is increased, the average Nusselt number is decreased since the flow in the narrow section of annulus is reduced progressively.

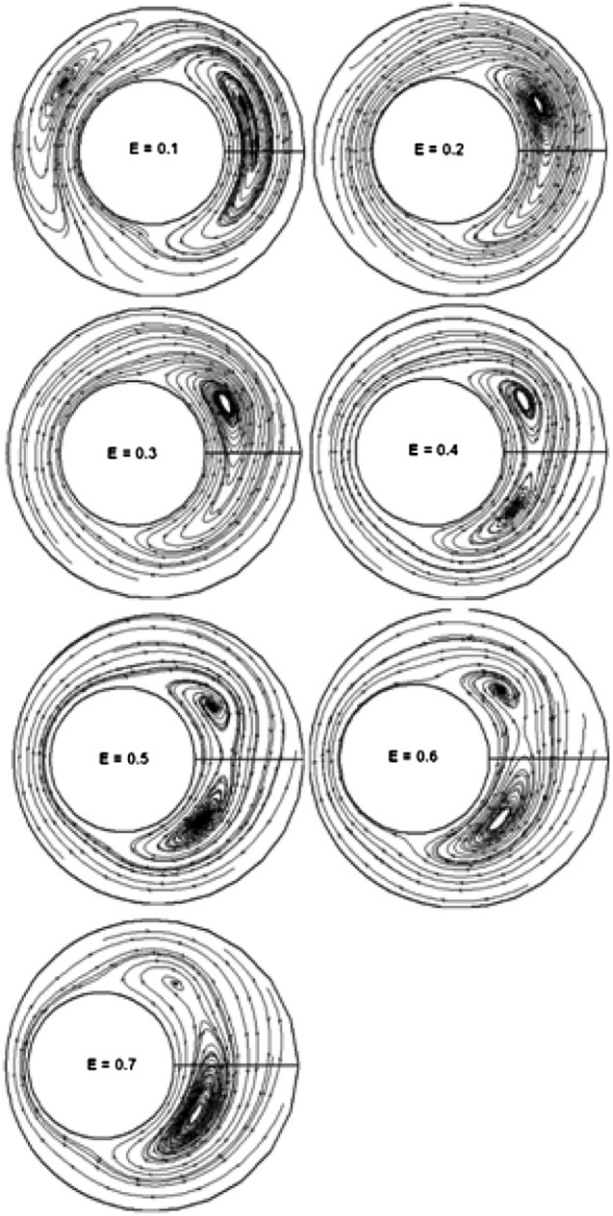


Fig. 15. Streamlines at $N = 0.5$ and $\varphi = 45$ for different eccentricities.

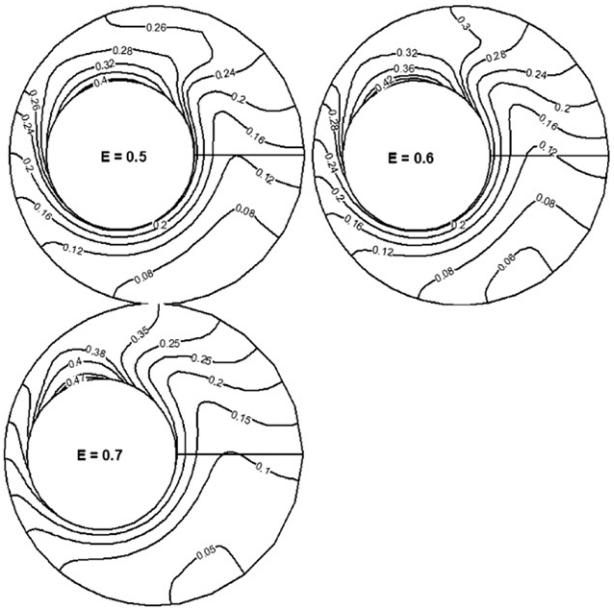


Fig. 16. Temperature contour lines at $N = 0.5$ and $\varphi = 45$ for different eccentricities.

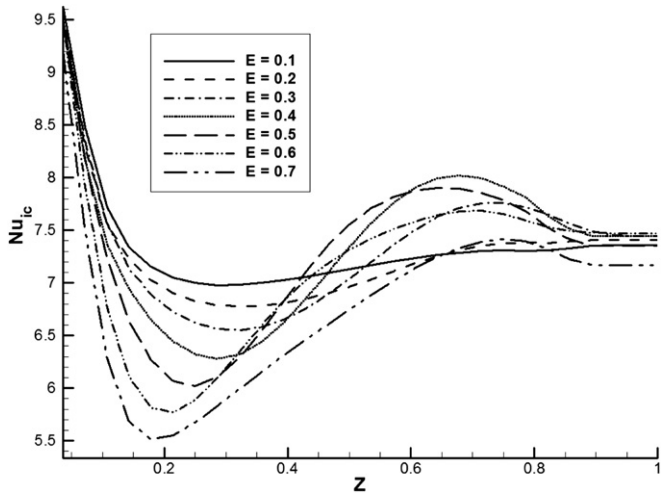


Fig. 17. Circumferential local Nusselt number of inner wall versus dimensionless length at $N = 0.5$ and $\varphi = 45$ for different eccentricities.

In Fig. 14, axial velocity contours for different values of eccentricities, at $\varphi = 45$ and $N = 0.5$ are shown. For $E \leq 0.5$, by increasing the eccentricity the circumferential flow intensifies and pushes the peak of the axial velocity profile clockwise. At higher values of eccentricity ($E \geq 0.6$), the viscous force in the narrow section becomes stronger, consequently the circumferential secondary flow gets weaker and the peak of the axial velocity profile moves counterclockwise. On the other hand, viscous force decreases in wide section by increasing the eccentricity and the axial velocity increases in this region. Fig. 15, demonstrates the streamlines for different values of eccentricity at $\varphi = 45$ and $N = 0.5$. At small values of eccentricity, since the annulus is almost symmetric, two vortices are formed. As the eccentricity become larger, due to the increased temperature gradient in the narrow section, the circumferential flow is formed. For $0.2 \leq E \leq 0.6$, although the viscous force in narrow section becomes stronger, the increase in fluid temperature, and the resulting buoyancy force in this part is dominant and the circumferential flow intensifies. For $E = 0.7$, the effect of viscous force overcomes the influence of buoyancy force, consequently the circumferential secondary flow gets weaker. On the other hand, the increase fluid temperature in the lower part of the annulus in the wide section, as shown in Fig. 16, would cause the lower vortex becomes dominant. For $0.4 \leq E \leq 0.5$, the circumferential flow becomes strong enough to split the vortex in two parts and weaken the upper one, as explained before.

In Fig. 17, variation of the circumferential local Nusselt number versus dimensionless length is plotted at the corresponding condition of Fig. 14. In the first stage, the Nusselt number decreases by increasing eccentricity due to thickening of the boundary layers. Further, for $E = 0.1$, since the circumferential secondary flow is weak, the Nusselt number varies slightly and remains almost constant. By increasing eccentricity in the range 0.2–0.6 ($0.2 \leq E \leq 0.6$), the circumferential secondary flow gets stronger and the axial velocity increases which results in increased local Nusselt number. Strengthening the secondary flow will result in increasing the Nusselt number in the initial length of the annulus. This shifts the maximum of the curve to the left. For $E = 0.7$, the circumferential secondary flow is weak enough to lose its effect on

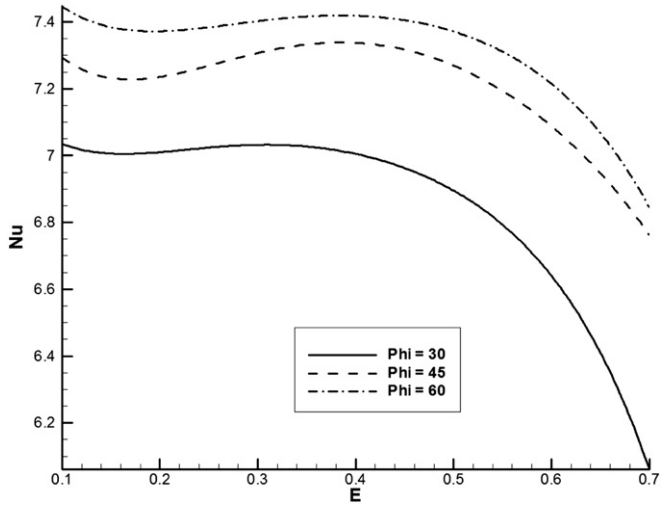


Fig. 18. Average Nusselt number of inner wall versus eccentricity at $N = 0.5$ for three different inclination angles.

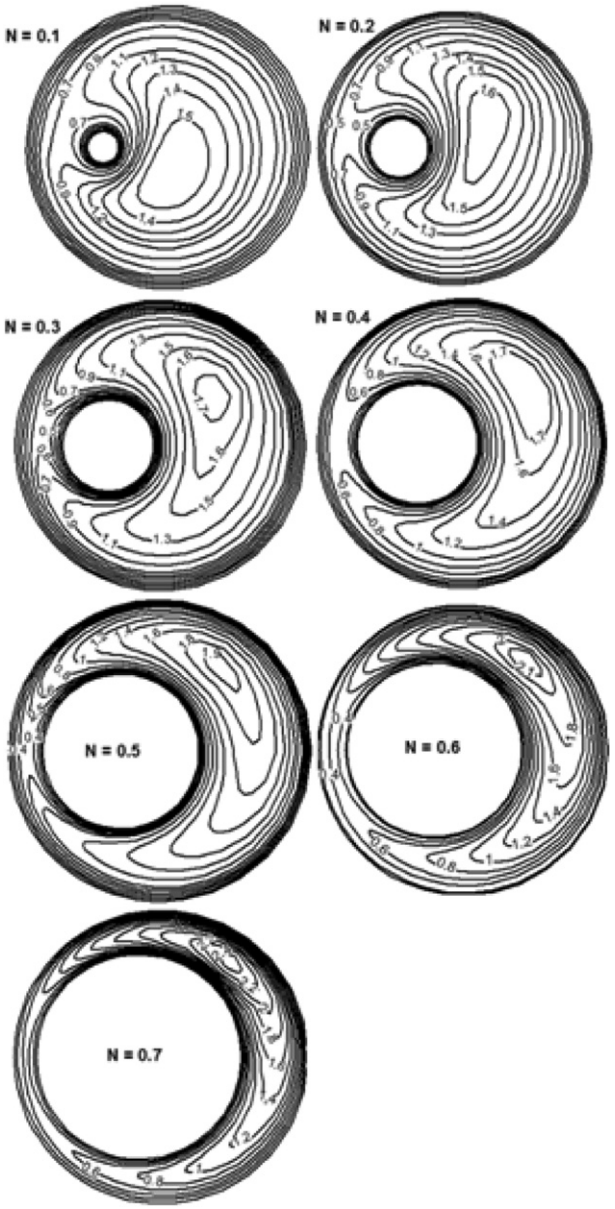


Fig. 19. Axial velocity contour lines at $E = 0.5$ and $\varphi = 45$ for different radius ratios.

variation of the local Nusselt number along the annuli. Consequently the third part of its development is delayed and the rise in the second part of the curve is shifted toward the end of the annulus.

Fig. 18 shows the average Nusselt number of inner wall versus eccentricity for different inclination angles at $N = 0.5$. As eccentricity increases to $E = 0.2$ the average Nusselt number reduces slightly, due to the decrease in axial velocity. For $0.2 < E \leq 0.4$, the circumferential flow gets stronger and the Nusselt number rises. Further increase of eccentricity ($E \geq 0.5$), increases the viscous force in the narrow section, decreasing the axial velocity near the inner wall. Hence, the average Nusselt number reduces profoundly. At a fixed eccentricity, laying the annulus down, decreases the axial velocity slightly, but simultaneously intensifies the secondary flow, which results in higher average Nusselt number.

The axial velocity contours and secondary flow field streamlines for different radius ratio at $\varphi = 45$ and $E = 0.5$ are demonstrated in Figs. 19 and 20 respectively. As radius ratio increases, in addition to enlarging the heat transfer area, the cross sectional area of annulus decreases, while the mass flow rate remains constant. As a result, the axial velocity increases. At small radius ratios, due to small heat transfer area, and hence low temperature of fluid, the secondary flows are weak. As radius ratio increases, the secondary flows get stronger up to $N = 0.5$. For larger radius ratios ($N > 0.5$), due to reduction of the cross sectional area and stronger viscous force, the secondary vortex flow becomes weak. On the other hand, the increase in the bulk temperature of fluid makes the circumferential flow stronger, which splits the vortex in to two, and the upper one gets weaker.

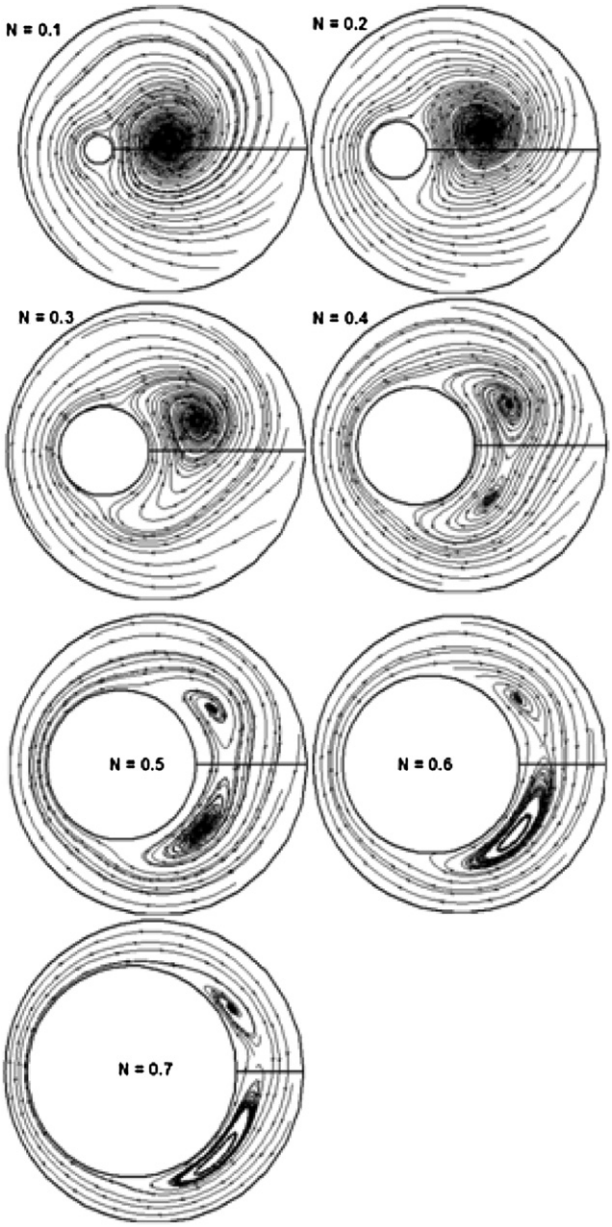


Fig. 20. Streamlines at $E = 0.5$ and $\varphi = 45$ for different radius ratios.

Fig. 21 shows the variation of the circumferential local Nusselt number of the inner wall versus the dimensionless length of annuli for different radius ratios. The trend is similar to Figs. 12 and 17, which were discussed earlier.

The average Nusselt number for the inner wall versus radius ratios for different angles of inclination at $E = 0.5$ are plotted in Fig. 22. As radius ratio increases the average Nusselt number decreases. The explanation of this effect can be stated by the fact that as the heat transfer area enlarges, the temperature gradient increases in the cross section plane, which increases the convection coefficient and the Nusselt number. Also by increasing the degree of inclination the average Nusselt number becomes larger, due to the effect of stronger secondary flow.

A function is developed in term of studied parameters to predict the Nusselt number value. It can be calculated using this relation with minimum error which can be expressed as

$$Nu = 0.3055 + 0.9157\sin\left(\frac{1.3935\pi\varphi}{180}\right) - 0.1187\exp\left(4.133E^2\right) + 3.013E^{0.1715} + 1.4905\exp\left(-0.0765N\right) + 1.084N^{-0.5882} \tag{38}$$

where its output is validated in following studied ranges: $10 \leq \varphi \leq 80$, $0.1 \leq E \leq 0.7$ and $0.1 \leq N \leq 0.7$.

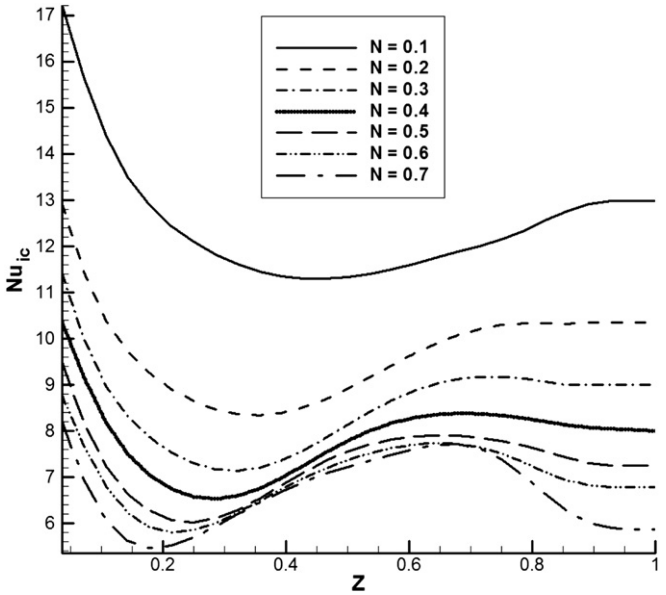


Fig. 21. Circumferential local Nusselt number of inner wall versus dimensionless length at $E = 0.5$ and $\varphi = 45$ for different radius ratios.

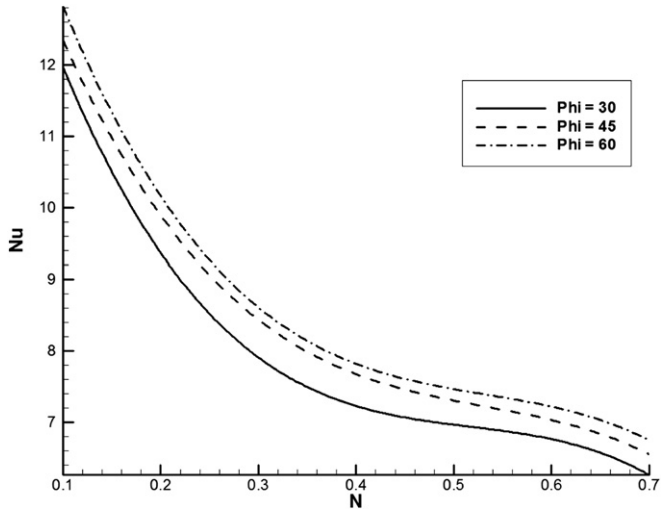


Fig. 22. Average Nusselt number of inner wall versus radius ratio at $E = 0.5$ for three different inclination angles.

7. Summary and conclusion

The flow field and heat transfer of laminar mixed convection in an inclined eccentric straight annulus are investigated numerically. The governing equations including the three-dimensional Navier–Stokes, the continuity and the energy equations are derived in cylindrical bipolar coordinate and are solved by using a second order finite difference method based on the Projection algorithm. Different values of eccentricities, inclination angles and radius ratios were studied. Developing hydrodynamic and thermal regimes was considered in the study. The inner wall is subjected to a fixed heat flux and the outer wall is insulated.

It is shown that as annulus is inclined down, the circumferential secondary flow pushes the peak of the axial velocity profile clockwise and splits the vortex. Also the average Nusselt number reaches its maximum and then decreases. Furthermore, the pattern of the axial velocity and secondary flows change as the eccentricity increases. In this case, the average Nusselt number again has a maximum. Finally, the results also show that the axial velocity increases despite the secondary flow becomes weaker and the average Nusselt number decreases by increasing the radius ratio. Also, a numerical correlation is developed to predict the Nusselt number value in term of φ, E and N under studied ranges.

Appendix A. Validation test of Boussinesq approximation

To check the validity of Boussinesq approximation, first properties of air at 300 K are obtained as following values:

$$\beta = 3.33 \times 10^{-3} \frac{1}{\mathrm{K}}, \quad k = 0.02227 \frac{\mathrm{W}}{\mathrm{m\,K}}, \quad \nu = 15.69 \times 10^{-6} \frac{\mathrm{m}^2}{\mathrm{s}}$$

In addition to gravitational acceleration and outer radius values:

$$g = 9.864 \frac{\mathrm{m}}{\mathrm{s}^2}, \quad r_\mathrm{o} = 0.1\ \mathrm{m}$$

According to Grashof number and its values in present work:

$$Gr = \frac{\beta g q' D_\mathrm{h}^4}{k\nu^2} = 1.35 \times 10^5$$

The amount of $q'D_\mathrm{h}/k$ can be calculated at each radius ratio. Appendix Table A1 shows values for $q'D_\mathrm{h}/k$ at different radius ratios and corresponding maximum dimensionless temperature (θ_max).

Appendix Table A1
Maximum dimensionless temperature at corresponding radius ratios.

N	0.1	0.2	0.3	0.4	0.5	0.6	0.7
$\frac{q'D_\mathrm{h}}{k}$	0.1733	0.2467	0.3683	0.5848	1.0107	1.9740	4.6791
θ_max	0.128	0.1967	0.270	0.348	0.471	0.576	0.735

where $\theta = T - T_\mathrm{e}/q'D_\mathrm{h}/k$.

Based on data in Appendix Table A1, maximum temperature difference can be calculated which is related to $N = 0.7$.

$$(T - T_\mathrm{e})_\mathrm{max} = 3.44\ \mathrm{K}$$

And it can be observed that the Boussinseq approximation is valid for this maximum temperature difference value.

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